RAGHVENDER TYAGI

7253020660 | raghvendertyagi@gmail.com
LinkedIn | LeetCode | GitHub

EDUCATION

KIET Group of Institutions, Ghaziabad *Oct 2023 – 2027*
B.Tech in Computer Science & Engineering (AI) | CGPA: 7.6

TECHNICAL SKILLS

Languages: Java, Python, SQL
Backend: Django, PostgreSQL, Redis, Celery, REST APIs
GenAI: LangGraph, LangChain, RAG, Model Context Protocol (MCP), AI Agents
Distributed Systems: Docker, WebSockets (Django Channels), Redis Pub/Sub, System Resource Monitoring
CS Fundamentals: DSA, OOP, Operating Systems, DBMS, Computer Networks

INTERNSHIP

Lightning Leap Analytics *Jan 2026 – May 2026*
AI Research Intern | Remote

- Contributed to Tequity, an AI-driven enterprise analytics platform enabling natural language query retrieval, semantic metadata indexing, and automated business intelligence across structured and unstructured datasets.
- Built an agentic multi-agent workflow using LangGraph featuring state persistence, custom tool execution, and dynamic query routing between task-specific sub-agents.
- Implemented a hybrid Text-to-SQL retrieval engine backed by PostgreSQL (pgvector) and a dual-layer AST/regex validator (validate_sql_access) to enforce role-based access control (RBAC) and prevent unauthorized raw data exposure.

LIVE PROJECTS

TakeMyCompute | Distributed P2P Compute Marketplace
Tech Stack: Django, Django Channels, Redis, Docker, PostgreSQL, Celery | Live Link

- Built a distributed cloud compute platform enabling providers to monetize idle hardware by executing renters' workloads in secure, sandboxed Docker environments.
- Designed a lightweight Python provider agent using psutil to report live system metrics (CPU/RAM/Disk) via WebSockets on periodic heartbeats.
- Implemented Docker-based task isolation via docker-py with strict cgroup limits (CPU/RAM caps) and isolated bridge networking to prevent host abuse.
- Architected a real-time bidirectional pipeline with Django Channels and Redis Pub/Sub for low-latency container log streaming, backed by automatic reconnection and REST fallback for zero data loss.

Paperly AI | Autonomous AI Research Engine
Tech Stack: Django, LangGraph, Python Multi-Threading, Celery, Redis, OpenAI/Gemini API | Live Link

- Architected an agentic research pipeline leveraging LangGraph, executing parallel LLM workers via ThreadPoolExecutor to cut end-to-end generation time by 65%.
- Designed a scalable backend system using Django, Celery & Redis to handle long-running LLM tasks asynchronously, eliminating request timeouts in production.
- Automated literature analysis & novelty scoring by integrating the arXiv API, parsing abstract metadata to ground paper synthesis in real scientific literature.

AI-Powered GitHub Change Impact Analyzer

- Built a system that analyzes pull requests using dependency graphs, static analysis, and ML to flag risk-prone changes and affected modules.
- Surfaced context-aware insights to speed up code review and reduce the chance of production bugs from merged changes.

PATENTS

AI-Powered Robotic Guidance System for Visually Impaired Individuals *Application Filed, 2024*

- AI-driven robotic navigation system providing real-time environment perception and audio description of surroundings.

HACKATHONS & ACHIEVEMENTS

- INNOTECH Winner – KIET Group of Institutions, Ghaziabad
- OpenAI State Finals – Noida International University, Noida
- Techno-Business Finals – Jaipuria Institute of Management, Noida
- Innovation Project Award – Smart Irrigation System
- 1st Rank (Department Teams) – Smart India Hackathon (Institute-Level Selection)

CERTIFICATIONS

- AWS Certified AI Practitioner (AWS Training and Certification)
- LangGraph – LangChain Academy